

Hematology Final High-Yield Codex Guide

- Block: Hematology
- Method: Codex-authored synthesis from extracted lecture markdown and the batch guides in `codex_manual/`.
- Goal: concise final review, roughly 10 pages of markdown.

How To Think Through Heme

Most heme questions collapse if you decide which system is broken:

1. **RBC production:** marrow cannot make normal cells or hemoglobin.
2. **RBC survival:** cells are destroyed early, usually with reticulocytosis.
3. **Primary hemostasis:** platelet plug/vWF problem, mucocutaneous bleeding.
4. **Secondary hemostasis:** coagulation factor/fibrin problem, deep bleeding.
5. **Thrombosis:** clotting system tipped toward pathologic clot.
6. **Transfusion/cellular therapy:** right product, right compatibility, right immune toxicity.
7. **Lymphoid/plasma malignancy:** clinical tempo plus immunophenotype plus mutation/translocation.

RBC / Anemia / Hemolysis

Macrocytic / Megaloblastic Anemia

- **B12 deficiency:** megaloblastic anemia plus neurologic degeneration. Causes are usually malabsorption: intrinsic factor loss, gastric atrophy/gastrectomy, aging, acid suppression, pancreatic disease, terminal ileum disease.
- B12 clues: neuropathy, loss of vibration/position sense, cognitive/neuropsychiatric changes, low B12, elevated methylmalonic acid.
- **Folate deficiency:** poor diet, malabsorption, pregnancy/infancy, hemolytic anemia, drugs such as phenytoin/barbiturates/DHFR inhibitors.
- Folate deficiency is linked to neural tube defects.
- **Trap:** folate can improve anemia while B12 neurologic injury continues. Rule out B12 deficiency before folate-only treatment.

Hemolytic Anemia Framework

- Hemolysis is usually hyperproliferative: high reticulocytes unless marrow cannot respond.
- Congenital hemolytic anemias: **membranopathies, enzymopathies, hemoglobinopathies.**
- Intravascular hemolysis: low haptoglobin, high LDH, hemoglobinuria, plasma free hemoglobin.
- Extravascular hemolysis: splenic/liver macrophage clearance, jaundice, splenomegaly, pigmented gallstones.

G6PD vs PK vs Hereditary Spherocytosis

Disorder	Mechanism	Clues	Testing/Treatment
G6PD deficiency	X-linked low NADPH/glutathione protection	Episodic hemolysis after sulfa drugs, antimalarials, fava beans, infection; Heinz bodies -> bite/blister cells	G6PD activity, but repeat after crisis if normal; avoid triggers
Pyruvate kinase deficiency	AR low ATP from glycolysis defect	Chronic hemolysis from birth, neonatal jaundice, acanthocytes, high 2,3-DPG	Supportive, phototherapy, transfusion/EPO/splenectomy if severe
Hereditary spherocytosis	Membrane/cytoskeleton tethering defect	Spherocytes, high MCHC, splenomegaly, gallstones, congenital	Increased osmotic fragility, DAT negative; splenectomy if severe

- G6PD is episodic because oxidant stress is needed.
- PK deficiency is chronic because ATP failure is constant.
- HS and warm AIHA both cause spherocytes; **DAT negative = HS, DAT positive = warm AIHA.**

PNH

- Acquired PIGA mutation in hematopoietic stem cell -> loss of GPI anchors -> loss of CD55/CD59.
- CD55/CD59 normally protect against complement; loss causes complement-mediated intravascular hemolysis.

- Pattern: dark urine/hemoglobinuria, very high LDH, undetectable haptoglobin, DAT negative, no classic smear finding.
- Free hemoglobin depletes NO -> dysphagia, abdominal pain, erectile dysfunction, renal pigment injury, thrombosis.
- Diagnosis: flow cytometry showing CD55/CD59-negative RBCs and WBCs.
- Treatment: eculizumab anti-C5; vaccinate for Neisseria.

Sickle Cell Disease

- HbS = beta-globin point mutation on chromosome 11, glutamic acid -> valine.
- Deoxygenated HbS polymerizes -> sickling, rigidity, hemolysis, vaso-occlusion.
- HbAS trait: HbA about 60%, HbS about 40%, usually asymptomatic.
- HbSS/S beta-zero: HbS >90%, no HbA, severe.
- HbSC and S beta-plus are usually less severe but still sickle.
- Symptoms start in mid/late infancy as HbF falls.
- Acute complications: pain crisis/dactylitis, stroke, acute chest syndrome, splenic sequestration.
- Acute chest syndrome cycle: lung vaso-occlusion -> hypoxia -> deoxygenated HbS -> more sickling.
- Chronic: splenic autoinfarction, Howell-Jolly bodies, encapsulated organism risk; children get penicillin prophylaxis.
- Severe acute treatment: hydration, oxygen, exchange transfusion.
- Chronic treatment: hydroxyurea increases HbF and reduces HbS polymerization.

Sideroblastic Anemia, Lead, Porphyrias

- Sideroblastic anemia = failed heme synthesis -> iron trapped in erythroblast mitochondria -> ring sideroblasts.
- X-linked sideroblastic anemia: ALAS-2 mutation; B6 is ALAS-2 cofactor, so pyridoxine may help.
- Lead blocks ALA -> PBG and protoporphyrin -> heme; causes sideroblastic anemia, basophilic stippling, and neurotoxicity.
- Hepatic porphyrias use ALAS-1, not ALAS-2, so they cause neurocutaneous/liver symptoms rather than anemia.
- AIP: AD with incomplete penetrance, high ALA/PBG during attacks, abdominal pain, weakness, neuropsychiatric symptoms; triggers include alcohol, progesterone, rifampin, antiepileptics. Treat attacks with hemin.
- PCT: chronic blistering photosensitivity and liver disease; associated with alcohol and hepatic iron overload; treat with alcohol avoidance and phlebotomy.

Hemostasis / Coagulation / Platelets

Primary Hemostasis

Sequence:

1. Endothelial injury releases endothelin and vWF.
2. vWF binds collagen and platelet GPIb-IX-V.
3. Platelets activate, flip phospholipid, degranulate.
4. Dense granules release ADP/Ca²⁺; platelets release TXA₂.
5. ADP reveals GPIIb/IIIa.
6. Fibrinogen cross-links GPIIb/IIIa between platelets.

Key pairs:

- **vWF-GPIb = adhesion**
- **Fibrinogen-GPIIb/IIIa = aggregation**
- **ADP-P2Y₁₂ = activation/aggregation amplifier**

Primary vs Secondary Bleeding

Type	Components	Bleeding pattern
Primary hemostasis	Vessel wall, vWF, platelets	Petechiae, purpura, ecchymoses, epistaxis, gingival bleeding, menorrhagia
Secondary hemostasis	Tissue factor, coag factors, phospholipid, Ca ²⁺	Hemarthroses, muscle bleeds, deep hematomas, delayed bleeding

Platelet Disorders

- Thrombocytopenia severity: >50k often asymptomatic/procedure-safe; 20-50k bleeding with trauma/procedures; <20k spontaneous hemorrhage risk.
- ITP: IgG against GPIIb/IIIa -> splenic clearance; isolated thrombocytopenia, large platelets, increased megakaryocytes. Treat with steroids, IVIG, rituximab, TPO agonists, splenectomy.
- HIT: thrombocytopenia while on heparin plus thrombosis risk. Stop heparin and give non-heparin anticoagulant.
- Qualitative platelet disorder: mucocutaneous bleeding with normal platelet count, normal PT/aPTT.

Disorder	Defect	Clues
Bernard-Soulier	GPIb-IX-V adhesion defect	Large platelets, moderate thrombocytopenia, absent ristocetin response
Glanzmann thrombasthenia	GPIIb/IIIa aggregation defect	Normal count/morphology, poor ADP/epi aggregation, normal ristocetin
Storage pool disorder	Granule release/content defect	Impaired secondary aggregation

Coagulation Cascade

Memorize:

- Common pathway: **X, V, II, I**, then XIII stabilizes.
- PT/extrinsic: **tissue factor + VII**.
- aPTT/intrinsic: **XII, XI, IX, VIII**.
- Vitamin K-dependent: **II, VII, IX, X, protein C, protein S**.

Core flow:

- TF/VIIa activates X.
- VIIIa/IXa intrinsic tenase activates X.
- Xa/Va prothrombinase converts II -> IIa thrombin.
- Thrombin converts fibrinogen I -> fibrin Ia, activates XIII, activates platelets, and amplifies upstream factors.
- XIIIa cross-links fibrin.

Coagulation Disorders and Tests

- Hemophilia A = factor VIII deficiency; hemophilia B = factor IX deficiency.
- Hemophilia: X-linked, deep/joint/muscle bleeding, normal PT, prolonged aPTT, correction on mixing.
- Hemophilia A treatment: factor VIII, DDAVP, emicizumab. Hemophilia B: factor IX.
- vWD: most common inherited bleeding disorder; vWF mediates platelet adhesion and preserves factor VIII.
- vWD testing: vWF antigen, ristocetin cofactor activity, factor VIII level.
- Severe vWD can prolong aPTT because factor VIII falls.
- Vitamin K deficiency: prolonged PT and aPTT, corrects with mixing; treat vitamin K.
- Liver disease: all factors low except VIII/vWF; PT and aPTT prolonged, corrects with mixing.
- Mixing study: **correction = deficiency, no correction = inhibitor**.

Fibrinolysis and Natural Anticoagulants

- Plasmin breaks down fibrin; tPA converts plasminogen to plasmin.
- Alpha-2-antiplasmin inhibits plasmin; PAI-1 inhibits tPA.
- D-dimer reflects cross-linked fibrin breakdown.
- Antithrombin inhibits IIa, IXa, Xa, XIa.
- Protein C/S inhibit Va and VIIIa.

Thrombosis / Anticoagulants / TMA

Arterial vs Venous

- Arterial thrombosis: high pressure, plaque rupture, platelet-rich white thrombus.
- Venous thrombosis: low pressure, deep leg veins, fibrin/RBC-rich red thrombus.
- Virchow triad: stasis, endothelial injury, hypercoagulability.
- DVT symptoms: swelling from obstruction; pain/warmth/erythema from inflammation.

Thrombophilias

Disorder	Mechanism	Testing clue
Antithrombin deficiency	Less inhibition of IIa/IXa/Xa/XIa	Antithrombin activity
Protein C/S deficiency	Less inactivation of Va/VIIIa	Protein C/S activity
Factor V Leiden	Factor V resistant to activated protein C	F5 genetic test or APC resistance
Prothrombin G20210A	Increased prothrombin level	F2 genetic test
APS	Autoimmune thrombophilia	DRVVT, anticardiolipin, beta-2 glycoprotein

- APS paradox: prolonged phospholipid-dependent clotting tests in vitro, thrombosis/pregnancy morbidity in vivo.
- DRVVT in APS: prolonged, does not correct with mixing, corrects with excess phospholipid.

Anticoagulants

Drug/class	Target	Monitoring/Reversal
UFH	Antithrombin-dependent IIa and Xa inhibition	aPTT; protamine; HIT risk
LMWH	Antithrombin-dependent Xa inhibition	Usually no aPTT; protamine
Warfarin	Vitamin K epoxide reductase	INR; reverse vitamin K/FFP
Xa inhibitors	Direct Xa inhibition	No routine monitoring
Direct thrombin inhibitors	Direct IIa inhibition	Not antithrombin-dependent
tPA	Plasminogen → plasmin	Thrombolysis; bleeding risk
TXA/aminocaproic acid	Antifibrinolysis	Preserve clot in hyperfibrinolysis

- Warfarin initially lowers protein C/S faster than II/VII/IX/X, causing transient hypercoagulability. Bridge with heparin for at least 5 days.
- Warfarin skin necrosis = acquired protein C deficiency in skin/subcutaneous tissue.

TTP / HUS / DIC

- TMA = platelet-rich microthrombi shear RBCs → schistocytes, low platelets, intravascular hemolysis.

Syndrome	Mechanism	Labs/clinical	Treatment
TTP	ADAMTS13 deficiency → ultra-large vWF → platelet microthrombi	MAHA, severe thrombocytopenia, CNS, fever, mild renal, normal PT/aPTT, ADAMTS13 <10%	Plasma exchange, steroids/rituximab, caplacizumab
HUS	Shiga toxin endothelial injury	Bloody diarrhea, child, renal injury, normal PT/aPTT	Supportive, dialysis
DIC	Systemic tissue factor activation	Bleeding + thrombosis, low fibrinogen, high D-dimer, prolonged PT/aPTT	Treat underlying trigger

Transfusion / BMT / Cellular Therapy

Blood Products

Product	Use
RBCs	Severe anemia, often Hgb <7, acute blood loss
Platelets	<10k prophylaxis, <50k before invasive procedure
FFP	Multiple factor deficiencies: liver disease, DIC, warfarin
Cryoprecipitate	Low fibrinogen; contains fibrinogen, VIII, vWF, XIII

- Type A patient can receive A or O RBCs, and A or AB plasma.
- Type/screen/cross: ABO/Rh, antibody screen, donor-unit compatibility.
- Acute hemolytic transfusion reaction: preformed ABO antibodies, clerical error, fever/chills/hypotension/flank pain/hemoglobinuria/DIC. Stop transfusion, IV fluids, vasopressors.
- Delayed hemolytic reaction: anamnestic response to minor antigen after prior exposure.

BMT

- Autologous BMT: patient's own cells, no GVHD, useful for dose-escalation-sensitive tumors.
- Allogeneic BMT: donor immune system, GVHD risk, graft-vs-tumor benefit.
- GVHD organs: skin, GI tract, liver.
- GVHD is a double-edged sword: lower relapse but more toxicity.
- Reduced-intensity transplant lowers conditioning toxicity and relies on graft-vs-tumor.
- Post-transplant cyclophosphamide makes haploidentical transplant more feasible by deleting alloreactive T cells.

CAR-T / Bispecifics / CRS

- CAR-T: autologous engineered T cells, 2-4 week manufacturing, single infusion, potential durable persistence, higher CRS risk.
- Bispecific antibodies: off-the-shelf, redirect endogenous T cells, serial/continuous dosing, CRS often lower grade.
- Targets: CD19 for B-ALL/DLBCL, BCMA for myeloma, CD20 for B-NHL.
- CRS mechanism: T-cell activation -> cytokines -> macrophage IL-6-driven systemic inflammation.
- CRS management: grade 1 fever = supportive; grade 2 fever + hypotension = tocilizumab; grade 3/4 = tocilizumab + steroids.

Lymphoid / Plasma Cell Malignancy

Architecture and Tempo

- Reactive hyperplasia: polyclonal, architecture preserved, tingible body macrophages, mixed maturation.
- Lymphoma: clonal/light-chain restricted, architecture effaced.
- Low-grade lymphomas: mature cells, slow, apoptosis escape, often incurable but controllable.
- High-grade lymphomas: rapid proliferation, B symptoms, high LDH/uric acid, high Ki-67, tumor lysis risk, often chemo-sensitive.

Low-Grade B-Cell Lymphomas

Disease	Anchor	Clinical pattern
Follicular lymphoma	t(14;18), BCL2, CD5-, CD10+, CD20+	Painless generalized LAD, watch-and-wait if asymptomatic
CLL/SLL	CD5+, CD19+, CD20+, CD23+	Older adult, lymphocytosis/smudge cells, hypogammaglobulinemia, AIHA
Mantle cell	t(11;14), cyclin D1, CD5+, CD23-	Older men, disseminated, GI/marrow/spleen, more aggressive
Marginal zone/MALT	CD5-, CD10-, CD20+	Chronic antigen stimulation: H. pylori, Sjogren, Hashimoto, HCV
Hairy cell leukemia	BRAF V600E, CD11c/CD25/CD103	Cytopenias, splenomegaly, dry tap

- CLL poor-risk: del17p/TP53, del11q/ATM, unmutated BCR genes.
- Treat CLL for progressive marrow failure, massive organomegaly, organ compromise, autoimmunity, or B symptoms.
- Richter transformation: CLL/SLL -> aggressive large B-cell lymphoma; rapidly enlarging nodes, B symptoms, high FDG uptake.

High-Grade and Hodgkin

Disease	Anchor	Clinical pattern
DLBCL	BCL6 common, CD20+	Most common NHL, rapidly enlarging mass, extranodal possible, R-CHOP, curable
Burkitt	MYC t(8;14), Ki-67 >95%, starry sky	Fastest tumor, jaw in endemic/EBV, abdominal in sporadic, tumor lysis risk
T-lymphoblastic lymphoma	TdT+ precursor T cells	Teen male, mediastinal mass, SVC syndrome
Classical Hodgkin	Reed-Sternberg, CD15+, CD30+, CD20-, CD45-	Painless cervical/supraclav LAD, mediastinal, contiguous spread, excellent prognosis
NLPHL	Popcorn cells, CD20+, CD15-, CD30-	Indolent, localized, can transform to DLBCL

Disease	Anchor	Clinical pattern
ATL	HTLV-1 CD4 T-cell malignancy	Skin lesions, lytic bone, hypercalcemia, flower cells

- Hodgkin mass is mostly inflammatory background; Reed-Sternberg cells are rare but diagnostic.
- HL B symptoms: fever, drenching night sweats, weight loss. Spleen is lymphatic tissue, not stage IV by itself.
- Classical HL subtypes: nodular sclerosis most common/mediastinal; mixed cellularity EBV+; lymphocyte-rich best prognosis; lymphocyte-depleted aggressive/older/HIV.

Plasma Cell Disorders

- Workup: SPEP, immunofixation, quantitative immunoglobulins, serum free light chains, UPEP, CBC/CMP, imaging, smear for rouleaux, marrow biopsy/cytogenetics.
- Suspect myeloma: macrocytic anemia, renal failure, pain/lytic lesions, protein gap, proteinuria, hypercalcemia.
- CRAB: hyperCalcemia, Renal failure, Anemia, Bone lesions.

Disorder	Monoclonal protein	Marrow plasma cells	CRAB
MGUS	Yes	<10% if checked	No
Smoldering myeloma	Yes	>10%	No
Multiple myeloma	Yes	>10%	Yes/end-organ damage

- MGUS progression risk about 1% per year; risk factors are non-IgG, M protein >1.5 g/dL, abnormal free light-chain ratio.
- Myeloma treatment is lifelong: transplant eligible often induction -> autologous SCT -> maintenance; common regimens include VRd and Dara-VRd.
- Supportive bone care: zoledronic acid or denosumab.
- IgM/Waldenstrom hyperviscosity: headache, blurry vision, epistaxis, dyspnea, bleeding. Treat urgent hyperviscosity with plasma exchange before rituximab.

Final Must-Not-Miss Traps

- Folate can hide B12 anemia while neurologic injury worsens.
- G6PD testing can be falsely normal during acute hemolysis.
- HS and warm AIHA both have spherocytes; DAT separates them.
- Sickle solubility detects HbS but does not distinguish trait from disease.
- vWF is adhesion and factor VIII stabilization.
- GPIb is adhesion; GPIIb/IIIa is aggregation.
- PT/aPTT do not assess platelet function.
- Mixing study correction = deficiency; no correction = inhibitor.
- APS prolongs tests but causes thrombosis.
- Warfarin starts procoagulant; bridge it.
- HIT means thrombosis risk; stop heparin and anticoagulate.
- TTP/HUS usually have normal PT/aPTT; DIC prolongs them and lowers fibrinogen.
- RBC compatibility and plasma compatibility run in opposite directions.
- Low-grade lymphoma can transform; high-grade lymphoma can be curable.
- CLL and mantle cell are both CD5+; CD23 and cyclin D1 separate them.
- Myeloma requires end-organ damage; many older patients have MGUS.